

Portfolio 5: Strings

Password Strength Analyzer

This program checks whether the user's password is weak, medium, or strong.

The program uses:

- Strings
- String methods
- Conditional statements
- Loops
- User input

The password will be checked based on:

- Length
- Uppercase letters
- Lowercase letters
- Numbers
- Special characters

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```
In [3]: # Password Strength Analyzer

print("=====")
print("      PASSWORD STRENGTH ANALYZER")
print("=====")

# Ask user for password
password = input("Enter your password: ")
```

```
=====
      PASSWORD STRENGTH ANALYZER
=====
```

```
In [4]: # Loop through every character in password

for char in password:

    # Check uppercase Letters
    if char.isupper():
        has_upper = True

    # Check Lowercase Letters
    elif char.islower():
        has_lower = True
```

```
# Check numbers
elif char.isdigit():
    has_number = True

# Check special characters
else:
    has_special = True
```

In [5]: *# Variables for checking password contents*

```
has_upper = False
has_lower = False
has_number = False
has_special = False
```

In [6]: *# Count password score*

```
score = 0

# Check password length
if len(password) >= 8:
    score += 1

# Check other conditions
if has_upper:
    score += 1

if has_lower:
    score += 1

if has_number:
    score += 1

if has_special:
    score += 1
```

In [7]: *# Display password analysis*

```
print("      PASSWORD ANALYSIS")

print("Password Length:", len(password))

print("Contains Uppercase:", has_upper)
print("Contains Lowercase:", has_lower)
print("Contains Number:", has_number)
print("Contains Special Character:", has_special)
```

PASSWORD ANALYSIS

Password Length: 9

Contains Uppercase: True

Contains Lowercase: True

Contains Number: False

Contains Special Character: True

```
In [8]: # Determine password strength

print("Password Strength Result:")

if score <= 2:
    print("Weak Password")
elif score == 3 or score == 4:
    print("Medium Password")
else:
    print("Strong Password")
```

Password Strength Result:

Medium Password